3.9 Inventing a New Function

WORKSHEET

The question

STEP II 2021, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does it actually mean to "solve" an equation? Is $x = \sqrt{2}$ a solution of $x^2 = 2$, or just a name for the solution?

STAGE 1 The curve $y = xe^x$

Covers: Part (i)

Ask yourself Differentiate xe^x. Where is the stationary point, and what kind is it?

Ask yourself What happens to xe^x as $x \to +\infty$ and as $x \to -\infty$? Where does the curve cross the axes?

STAGE 2 The inverse

Covers: Part (ii)

Pause and try For an inverse, f must be one-to-one on $[a, \infty)$. What is the smallest a that works? Sketch $y = g(x)$ by reflecting in $y = x$.

STAGE 3 Four equations

Covers: Part (iii)

Ask yourself The defining property is $g(z)\, e^{g(z)} = z$. In every part, aim for (something) $e^{(\text{the same something})} =$ (a number). For (a), multiply by e^x.

Ask yourself For (b) and (c), the unknown appears as $x \ln x$. Which substitution turns $x \ln x$ into something of the form ue^u?

Pause and try Remember the lowest value of ue^u. For (c), how many solutions does $ue^u = -\frac{1}{3}$ have, and which one does g give? For (d), exponentiate and aim for a multiple of x in the exponent.

STAGE 4 $x^x = 10$

Covers: Part (iv)

Ask yourself Take logarithms, then use the same substitution as in (b) and (c).

What's really going on

A function with a long history and a recent name

Johann Heinrich Lambert studied equations of this type in 1758, and Euler took them further in 1783. But the function had no standard name until the 1990s, when the developers of the computer algebra system Maple needed one. In 1996 Robert Corless, Gaston Gonnet, David Hare, David Jeffrey and Donald Knuth published a paper, "On the Lambert W function", that collected its properties and applications. It has been cited thousands of times, and W is now built into most mathematical software. The function g in the question is the *principal branch* W_0; the dashed curve in Figure 3.9.1 (in the full section online), which gave the second roots in (c) and (d), is the other real branch W_{-1}.

Where W appears

- **Wien's displacement law.** The wavelength at which a hot body radiates most strongly is inversely proportional to its temperature. The constant comes from solving $5(1 - e^{-x}) = x$, whose positive solution is $x = 5 + W(-5e^{-5}) \approx 4.965$. That is why the Sun, at about 5800 K, looks yellow-white, and why a cooler red dwarf looks red.
- **Power towers.** The infinite tower $x^{x^{x^{\cdots}}}$ converges for $e^{-e} \le x \le e^{1/e}$, and its value is $\dfrac{W(-\ln x)}{-\ln x}$.
- **Population and epidemic models.** The final size of an epidemic in the simplest model, and the time to reach a given population in some growth models, are both given by W.

Elementary functions

Is W really new, or could it be written using exp, ln, roots and the trigonometric functions? It is genuinely new: a theorem going back to Joseph Liouville in the 1830s shows that W is not an *elementary function*. The same theory shows that some innocent-looking integrals, such as $\int e^{-x^2}\,\mathrm{d}x$, cannot be done in elementary terms, which is why the normal distribution needs its own special function.

Try this next

1 **Two, four, and one more.** The equation $x^2 = 2^x$ has the obvious solutions $x = 2$ and $x = 4$ (compare Section 3.5). Show that it also has a negative solution, and that this solution is

$$x = -\frac{2}{\ln 2}\,\mathrm{g}\!\left(\frac{\ln 2}{2}\right) \approx -0.767.$$

2 **A derivative.** By differentiating $\mathrm{g}(x)e^{\mathrm{g}(x)} = x$ implicitly, show that, for $x \neq 0$,

$$\mathrm{g}'(x) = \frac{\mathrm{g}(x)}{x\big(1+\mathrm{g}(x)\big)}.$$

What happens at $x = -\frac{1}{e}$?

3 **An integral.** Use the substitution $x = we^w$ to show that $\displaystyle\int_0^e \mathrm{g}(x)\,\mathrm{d}x = e - 1$.

4 **The tower.** Assuming the tower $T = \sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdots}}}$ converges, show that $T = \sqrt{2}^{T}$, and deduce that $T = 2$ or $T = 4$. Which is it? (Compute a few terms of the sequence $t_1 = \sqrt{2}$, $t_{n+1} = \sqrt{2}^{t_n}$, and think about the cobweb diagrams of Section 2.2.)

Further reading

- **Robert Corless et al., "On the Lambert W function"**, *Advances in Computational Mathematics* 5 (1996). The paper that named the function. The first sections are readable with this section behind you.
- **Brian Hayes, "Why W?"**, *American Scientist* 93 (2005). A lively essay on the function's history and uses. Free online.
- **Brian Conrad, "Impossibility theorems for elementary integration"**. Lecture notes, available online, explaining Liouville's theory. Advanced.

The full guided solution is in Section 3.9:
https://beyondstep.keithmcnulty.org/ch03/lambert-w.html